DS612 – Interactive Data Visualisation

Course Project Report

Understanding the Drivers of US Traffic Accident Severity

A Hypothesis-Driven Visual Analysis of the
US Accidents Dataset (2016–2022)

Group 7

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Abstract

This report presents a data-driven visual analysis of the US Accidents dataset (2016–2022), covering over 7.7 million accident records across 49 states. Using a hypothesis-driven, visualisation-centric methodology, we examine two primary hypotheses: (1) adverse weather conditions increase accident severity, and (2) rush hours increase accident severity.

Critically, the dataset encodes severity as the *impact on traffic flow* rather than physical injury severity—a distinction that fundamentally reshapes how insights must be interpreted.

The dataset was pre-processed with a 2×IQR outlier removal strategy on `Duration` and `Distance(mi)`, and 2023 data was excluded due to partial-year temporal skew, yielding 6.1 million clean records across 2016–2022.

Through Python-based visualisations backed by statistical tests, we validate, challenge, and extend both hypotheses, and derive two new hypotheses that emerge naturally from the severity definition itself.

1 Introduction

Road accidents represent a persistent public safety and urban planning challenge. Understanding what drives accident severity—and specifically what causes accidents to have prolonged traffic impact—enables smarter infrastructure design, better emergency response, and more effective policy. The US Accidents dataset, compiled by Moosavi et al. [1], provides one of the most comprehensive real-world accident records available for such analysis.

To frame our investigation, we follow the **CPQH** structure—Context, Problem, Questions, Hypothesis—a method that grounds our visual storytelling in facts, real issues, and investigative questions before forming claims.

CPQH Stage	**Our Application**
Context	US road accidents 2016–2022 across 49 states; severity scored 1–4 based on traffic disruption duration.
Problem	Traffic severity is not random. Environmental and temporal factors may systematically increase accident-induced traffic disruption.
Questions	Does adverse weather prolong accident-induced disruption? Do rush hours amplify severity? Does road infrastructure moderate severity?
Hypotheses	H1: Bad weather increases accident severity. H2: Rush hours increase accident severity. H3 & H4: Extended hypotheses derived from the severity definition.

Table 1. CPQH framework applied to this project

2 Dataset Overview

The dataset was sourced from Kaggle [1] and is backed by peer-reviewed academic publications, lending it strong reliability. Data was collected in real-time from multiple traffic APIs, government agencies, cameras, and road sensors.

Attribute	Value
Total Records	7,728,394
Features	46 original features
Full Time Range	February 2016 – March 2023
Analysis Range	2016 – 2022 (2023 excluded — partial year)
Coverage	49 US states
Licence	CC BY-NC-SA 4.0
Severity Scale	1 (minor impact) → 4 (major disruption)

Table 2. Dataset at a glance

Critical Note on Severity Definition

> *Severity in this dataset does **NOT** measure physical injury or fatality. It measures the duration and extent of traffic flow disruption caused by the accident. Severity 1 means a short delay; Severity 4 means significant, prolonged road blockage.*

This distinction is crucial: high-severity accidents are those that tie up traffic for longer, correlating with accident complexity, road type, response time, and infrastructure—not necessarily injury severity.

Reliability Justification

- Supported by peer-reviewed publications in ACM SIGSPATIAL (2019) [2].
- Multi-source data collection (traffic APIs, law enforcement, road sensors) eliminates single-source bias.
- Released under CC BY-NC-SA 4.0 with full provenance documentation.

Key Features Used

Feature	Type	Description
Severity	Ordinal (1–4)	Traffic flow impact level
Start_Time / End_Time	Datetime	Accident start and clearance time
Weather_Condition	Categorical	Raw weather text (grouped into 7 categories)
Sunrise_Sunset	Binary	Day or night at time of accident
Junction / Signal / Crossing	Boolean	Road infrastructure at accident location
Start_Lat / Start_Lng	Continuous	GPS coordinates for spatial mapping
State	Categorical	US state of occurrence
Distance(mi)	Continuous	Length of road extent affected
Duration_Min	Derived	End_Time − Start_Time in minutes

Table 3. Key features used in the analysis

3 Data Quality and Fault Analysis

The underlying dataset introduces structural biases and limitations that constrain the conclusions that can responsibly be drawn. These faults were identified through exploratory visualisation and must be considered when interpreting every chart in this report.

#	Fault	Primary Impact
1	Severity Label Imbalance	Severity 2 ≈80%; misleading raw severity ratios
2	Geographic Sampling Bias	CA over-represented; invalid cross-state comparison
3	Road Network Bias	Rural areas appear artificially safer
4	Weather Confounding	Weather–severity link is correlational, not causal
5	Exposure Bias (Time-based)	No normalisation by vehicles on road per hour
6	Temporal Data Drift	Year-on-year trends may reflect data artefacts
7	Missing Exposure Data	All severity ratios are accident-conditional
8	Lack of Contextual Features	No road type, speed limit, or traffic density

Table 4. Summary of identified data faults

3.1 Fault 1: Severity Label Imbalance

Severity 2 dominates (~80–85% of total data). Severity 3 and 4 are significantly underrepresented; Severity 1 is extremely rare. High-severity patterns become statistically weak due to low sample sizes.

Impact: All High Severity Rate percentages must be read in the context of their underlying counts. A small percentage-point shift can be substantial in absolute terms.

3.2 Fault 2: Geographic Sampling Bias

California has disproportionately high accident counts relative to its population share. Other large states (TX, FL) are significantly lower than expected. Coverage is dependent on reporting and sensor infrastructure, not actual accident rates.

Impact: Direct state-level comparisons are partially reflections of data collection coverage rather than true accident rates.

3.3 Fault 4: Confounding Variables in Weather Analysis

Weather does not occur in isolation. The analysis lacks simultaneous control for traffic density, vehicle speed, road type, and time of day.

Impact: Causal interpretations of H1 (weather → severity) must be treated as correlational

only.

3.4 Fault 5: Exposure Bias in Time-Based Analysis

No normalisation by number of vehicles on the road at each hour is applied. A 21% severity rate at 4 am with 1,000 accidents represents a fundamentally different risk level from 19% at 8 am with 50,000 accidents.

Impact: Rush-hour severity comparisons (H2) are misleading without exposure normalisation. All time-based conclusions are accident-conditional.

3.5 Fault 6: Temporal Data Drift

Accident counts increase year-on-year; the high-severity rate simultaneously declines sharply. This decline likely reflects expansion of API coverage (more Severity 2 records added), not genuine improvements in accident outcomes.

Impact: Year-over-year severity rate comparisons must be treated as exploratory. The 2023 partial year was excluded for this reason.

3.6 Fault 7: Missing Exposure Data (Critical)

The dataset contains no information on traffic volume, vehicle miles travelled, or number of active drivers. True risk per unit of exposure cannot be computed.

Impact: The most fundamental limitation for both H1 and H2. All severity ratios are *accident-conditional*, not *trip-conditional*.

> *While all visualisations accurately represent the data as collected, the dataset introduces multiple biases and structural limitations. Any conclusions drawn—including the validation of H1 and H2—must be qualified by these faults. The analysis demonstrates patterns and associations, not causal mechanisms. Deeper investigation using matched datasets (traffic volume, speed records, injury severity) would be required to move from correlation to causation.*

4 Data Preparation

Raw data was processed in Python (Pandas) through a multi-stage pipeline. After cleaning, 2×IQR outlier removal on `Duration` and `Distance(mi)`, and exclusion of 2023 data (only 3 months of records, creating temporal skew), the dataset yields **6,162,346 usable rows across 2016–2022**.

4.1 Preprocessing Pipeline

Stage	Action	Rationale
Column Selection	21 of 46 features retained	Focus on accident context, weather, temporal, and spatial signals
Missing Values	Median fill for numerics; 'Unknown' for categoricals; drop on critical fields	Preserve data volume while removing unreliable records
Datetime Parsing	`Start_Time` and `End_Time` parsed to datetime	Enables temporal feature extraction
Year Filter	2023 excluded	Only 3 months of 2023 records; partial year creates temporal skew
Outlier Removal	2×IQR on `Duration_Min` and `Distance(mi)`; Temperature: −30–130°F; Lat/Lng within CONUS bounds	Remove sensor errors and implausible values (e.g. 16-hour road closure, 8.5-mile single-accident road blockage)
Deduplication	Unique on accident ID	Prevent double-counting from multi-source collection

Table 5. Preprocessing pipeline summary

4.2 Engineered Features

Feature	Definition
`Severity_Group`	High (3–4) vs. Low (1–2) binary grouping
`Weather_Group`	7 categories: Clear, Cloudy, Rain, Snow/Ice, Fog/Haze, Wind/Dust, Other
`Road_Feature`	Junction / Signal / Crossing / Open Road (priority-ordered)
`Duration`	End_Time − Start_Time in minutes; clipped 0–1440, then 2×IQR outlier removal
`Accident_Count`	Flag that aggregates over categories to give accident count
`High_Severity_Ratio`	Ratio of High severity accidents over total accidents (considered High Severity as 1 and Low Severity as 0)

Table 6. Engineered features and their definitions

4.3 Weather Grouping Logic

The raw `Weather_Condition` field contains hundreds of distinct string values. These were mapped to 7 groups using keyword matching:

Group	Keywords matched (case-insensitive)
Snow/Ice	snow, sleet, ice, blizzard, wintry
Fog/Haze	fog, mist, haze, smoke, smog
Rain	rain, drizzle, shower, thunder, storm
Cloudy	cloud, overcast, partly
Clear	fair, clear, sunny
Wind/Dust	wind, dust, sand
Other	anything not matched above

Table 7. Keyword-based weather grouping rules

4.4 Rationale for Outlier Removal (2xIQR)

The 2×IQR method was chosen over standard 1.5×IQR to retain genuine long-duration incidents while removing sensor errors and timestamp mistakes. The distribution plots (V_B1, V_B2) directly validate this choice:

- Wyoming pre-clean maximum distance: ~8.5 miles (physically implausible for a single accident).
- Maine pre-clean maximum duration: ~950 minutes (~16 hours), almost certainly a timestamp entry error.
- Post-cleaning, the cross-state box shapes are consistent, confirming structural soundness of the cleaned dataset.

5 Hypotheses

Two primary hypotheses were formulated based on the dataset context, plus two extended hypotheses that emerge from the severity definition. Each is framed through task abstraction to map the analytical goal onto appropriate visual idioms.

ID	Type	Statement
H1	Primary	Bad weather conditions increase accident severity compared to normal conditions.
H2	Primary	Rush hours increase accident severity compared to other times of day.
H3	Extended	Road infrastructure with mandatory stopping controls (signals, crossings, stop signs) reduces severity more substantially than any single weather category increases it.
H4	Extended	Accidents during low-traffic periods (night, early morning, weekends) produce higher per-accident severity scores due to higher vehicle speeds and longer emergency response times.

Table 8. All hypotheses investigated in this report

5.1 H1 — Data and Task Abstraction

Data Abstraction: The dataset is a table where each row represents one accident. Key attributes: `Weather_Group` (categorical, 7 values), `Severity_Group` (binary: High/Low), `High_Sev_Rate` (quantitative ratio).

Task Abstraction: Compare the proportion of high-severity accidents (Severity 3–4) across different weather groups.
Problem breakdown: Compare → Category → Proportion.

5.2 H2 — Data and Task Abstraction

Data Abstraction: Key attributes: `Hour` (ordinal, 0–23), `DayOfWeek` (categorical), `High_Severity_Ratio` (quantitative).

Task Abstraction: Analyse whether the hourly distribution of accident severity peaks during rush hour windows (6–10 and 16–20).
Problem breakdown: Analyse → Temporal Trend → Identify Pattern.

6 Overview Visualisations

6.1 V1: Geographic Distribution of High-Severity Accidents

The dot map plots each accident by GPS coordinates, with Severity 3 in blue and Severity 4 in red. Low-severity accidents (1–2) are excluded to highlight the spatial concentration of disruptive incidents.

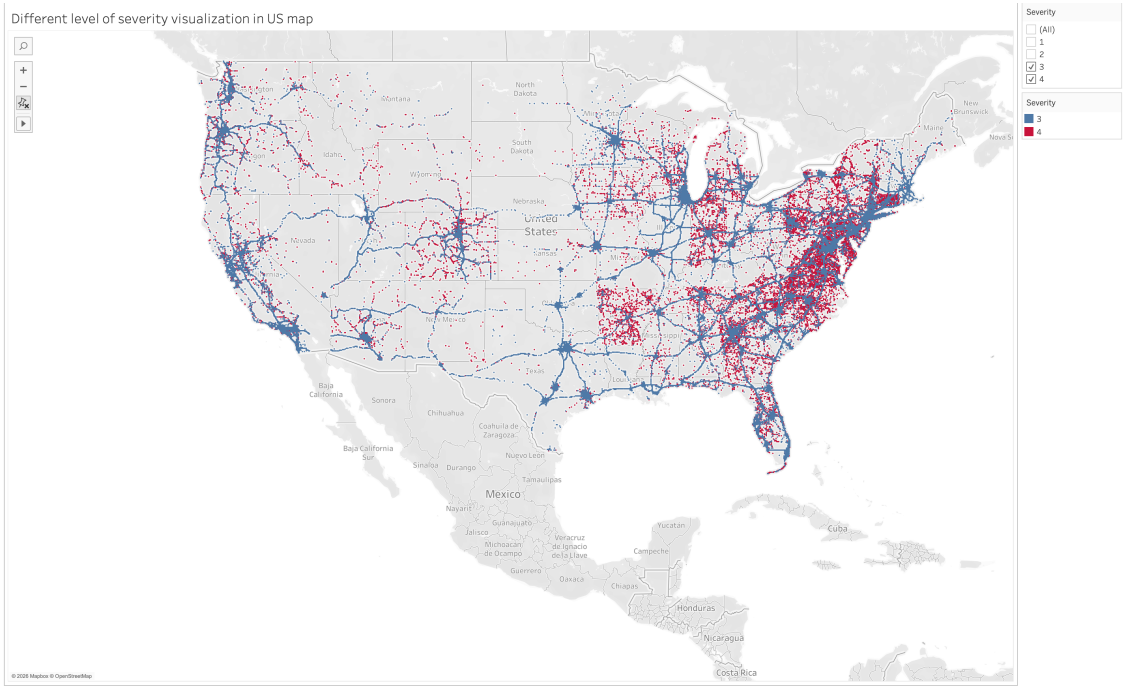


Figure 1. Geographic distribution of high-severity accidents (Severity 3–4)

Marks	0D Point
Channels	Position (ordered, unaligned) + Colour Hue (blue = Sev. 3, red = Sev. 4)
Action + Target	Identify spatial pattern in accident severity distribution
Scalability	500K points; overplotting managed by point size and opacity

Table 9. Analysis of idiom

Severity 4 accidents (red) cluster densely along the Eastern Seaboard and major interstate corridors, particularly in Florida, California, and the Texas triangle. The western interior shows predominantly Severity 3 (blue) in sparse highway networks. The geographic skew mirrors population density and road network complexity rather than environmental conditions alone.

6.2 V2: Top 25 States by Total Accident Count (2016–2022)

A ranked horizontal bar chart showing California leading with approximately 1.39M accidents, more than double Florida (687K) and three times Texas (454K). The sorted order enables immediate rank lookup and magnitude comparison.

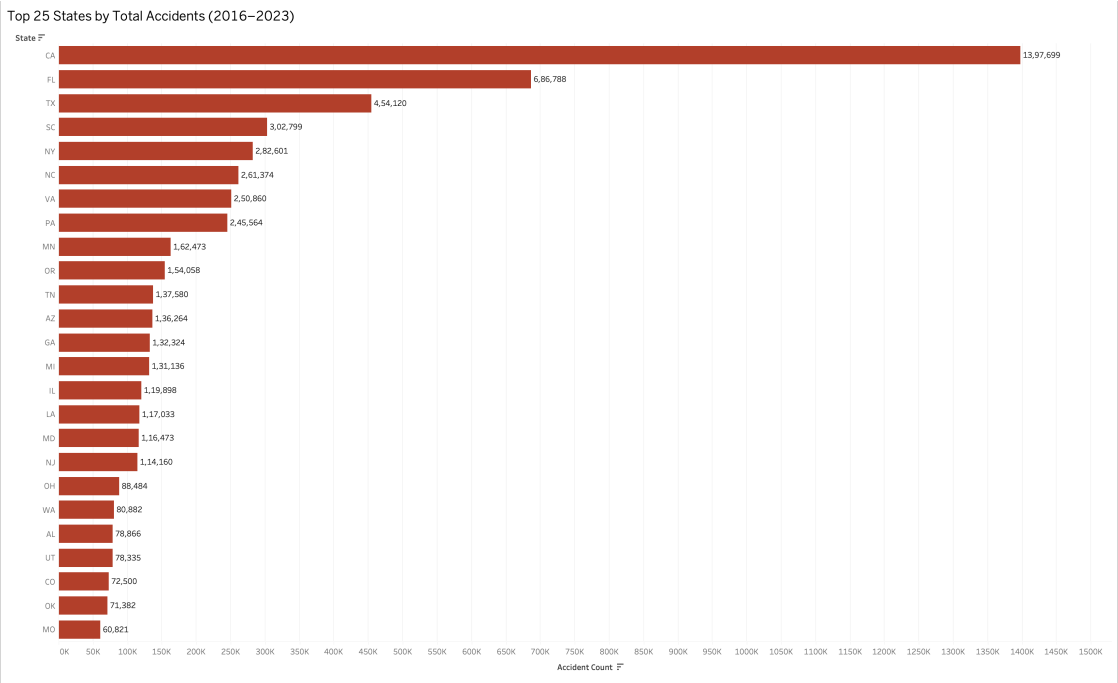


Figure 2. Top 25 states by total accident count

Marks	1D Line (bar)
Channels	Horizontal Length (ordered, aligned)
Action + Target	Lookup rank and compare magnitude across states
Scalability	Dozens of categories, thousands of values per bar

Table 10. Analysis of idiom

California's dominance is partly a data-collection artefact — the dataset relies on traffic API feeds, and California has one of the most comprehensive sensor networks in the US.

6.3 V3: Overall Severity Distribution

A simple bar chart reveals the sharp skew: Severity 2 dominates with 4.85M records, while Severity 1 (65K), Severity 3 (1.15M), and Severity 4 (103K) are comparatively small. This

distribution is critical for interpreting all downstream High Severity Rate calculations.

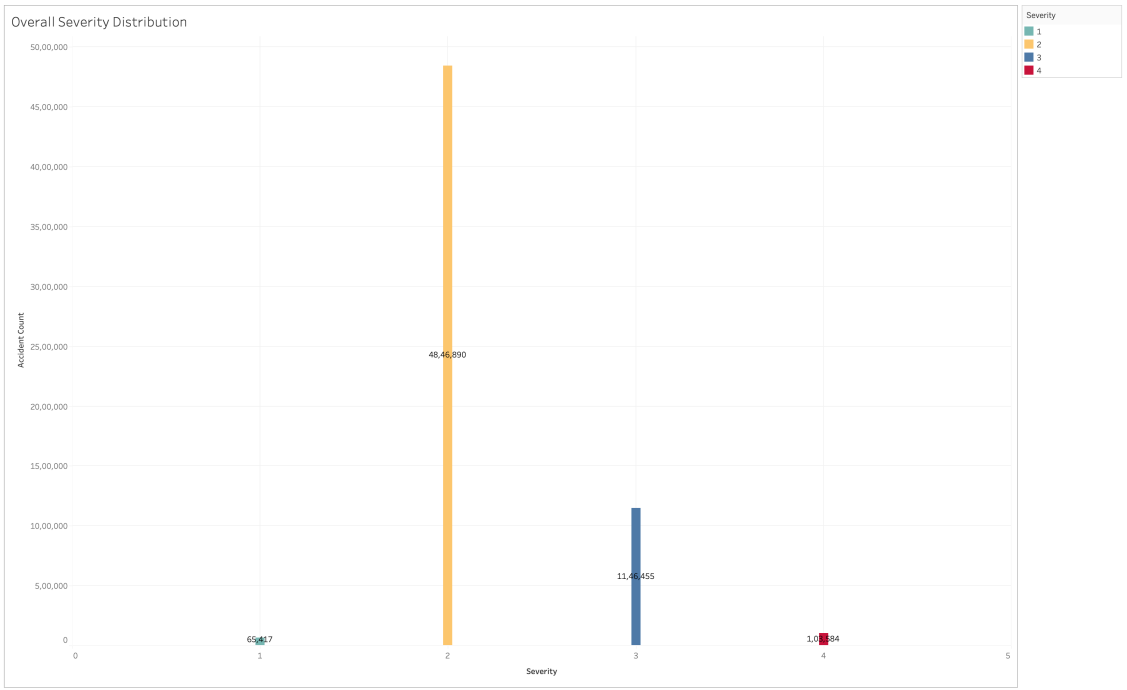


Figure 3. Overall severity distribution across the full dataset

Marks	1D Bar
Channels	Vertical Position + Colour Hue (categorical identity)
Action + Target	Compare counts across ordinal severity categories
Scalability	4 categories — optimal for grouped bars

Table 11. Analysis of idiom

Over 80% of accidents are Severity 2. This class imbalance means High Severity Ratio (proportion of Severity 3–4) is a more meaningful signal than raw counts throughout this report. A shift from 19% to 22% high severity represents a substantial change in absolute terms given the volume.

6.4 V4: Year-wise Accident Volume and High Severity Rate

A dual-axis combo chart pairs stacked bars (High vs. Low severity counts) with a dashed line for the average yearly severity rate. The COVID-19 period (2020–2021) is highlighted in pink.

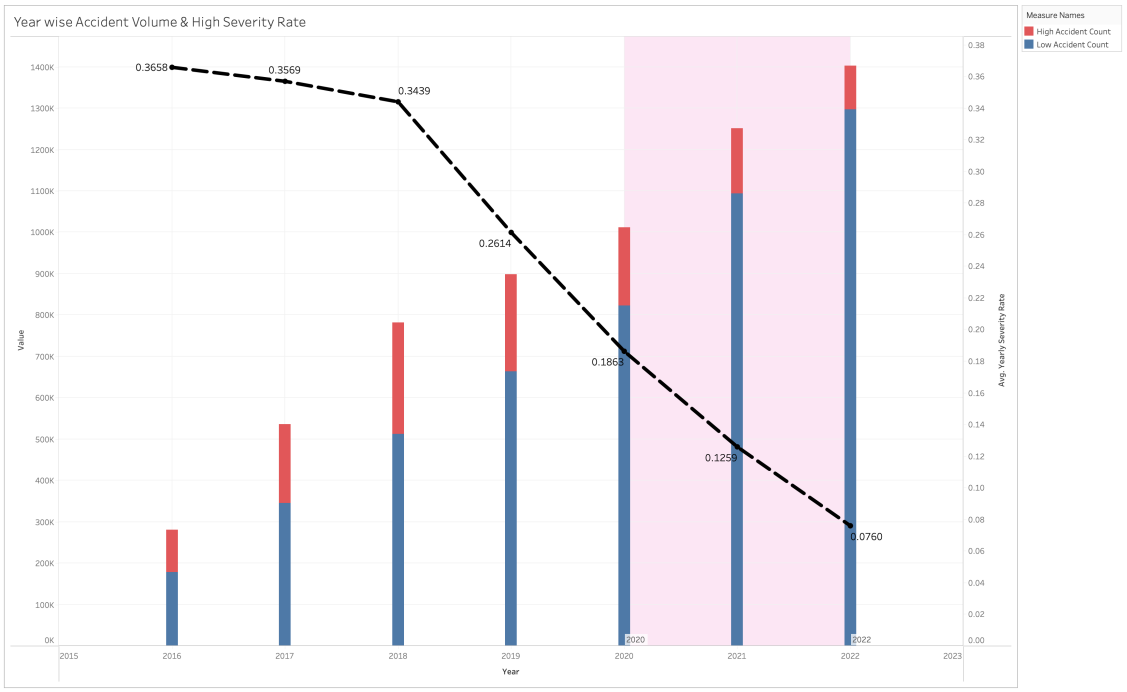


Figure 4. Year-wise accident volume (stacked bars) and high-severity rate (dashed line)

Marks	1D Bar (stacked) + 0D Point
Channels	Vertical Length + Colour Hue (stacked) + Position on secondary Y-axis (rate)
Action + Target	Compare trend over time; identify COVID anomaly
Scalability	7 years (2016–2022) — manageable temporal resolution

Table 12. Analysis of idiom

High severity rate declined sharply from 34% in 2016 to under 3% by 2022. This counter-intuitive long-term trend likely reflects the rapid growth in Severity 2 records as API coverage expanded year-on-year, diluting the high-severity proportion — not a genuine improvement in accident outcomes. The COVID dip in 2020 shows reduced accident volume but the rate continued its structural decline. Inter-year severity rate comparisons should be treated as exploratory, not conclusive.

7 Exploratory Distribution Analysis (Box Plots)

Prior to hypothesis testing, three 5-point summary box plots were generated to understand the distributional properties of key quantitative variables and validate the 2×IQR outlier removal strategy. These charts informed data cleaning decisions and reveal structural characteristics that qualify all downstream analysis.

7.1 V_B1: Accident Distance Distribution by State

A side-by-side box plot displays the distribution of `Distance(mi)` — the road length affected by the accident — categorized by all 49 states. Grey box plots show post-cleaning values; blue dotted whisker extensions show pre-cleaning outliers, making the effect of 2×IQR removal explicit.

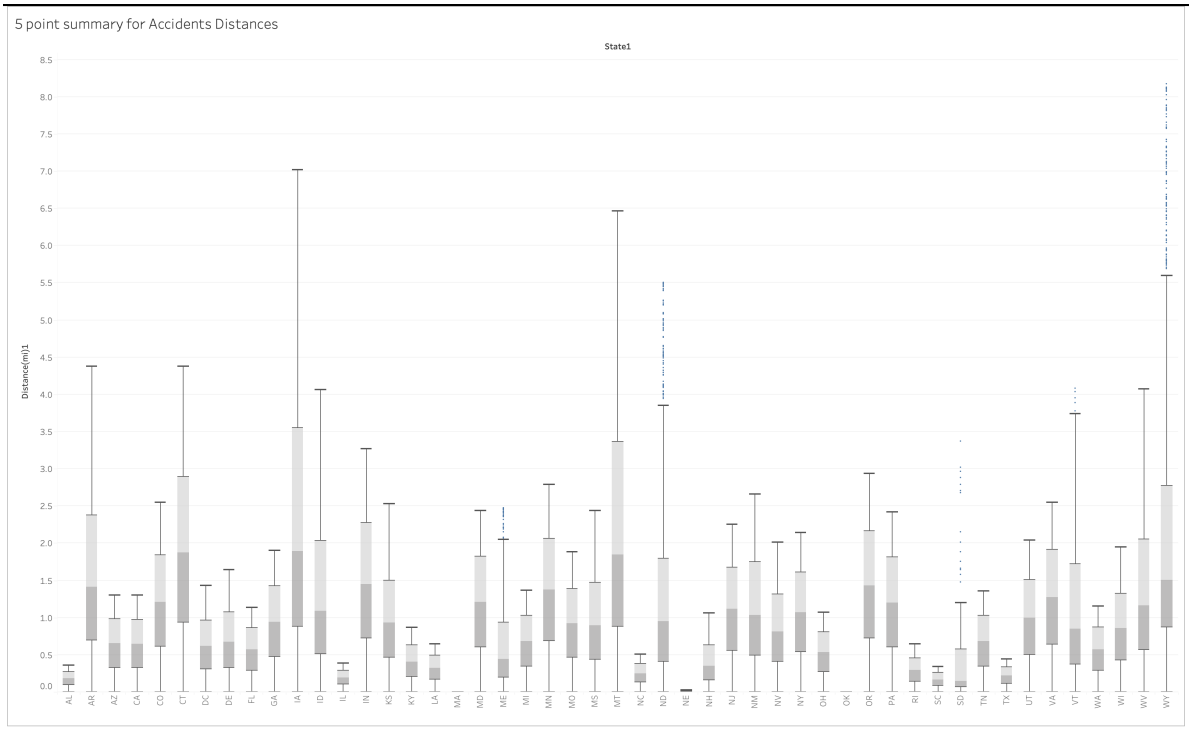


Figure 5. Accident distance distribution by state after 2xIQR cleaning

Marks	1D Line (box: Q1, median, Q3) + 0D Points (whiskers, outliers)
Channels	Vertical Length (ordered, aligned)
Action + Target	Analyse distribution spread; compare outlier removal effect across states
Scalability	49 states — manageable for a side-by-side layout with small font labels

Table 13. Analysis of idiom

Most states show a median accident distance of 0.5–1.5 miles (the IQR box), confirming the dataset predominantly captures localised road blockages. Georgia (GA) is the most notable outlier with max whisker ≈7 mi; Montana (MT) and Wyoming (WY) show wide IQR spreads reflecting sparse rural highway networks where accidents affect longer road stretches. Wyoming's pre-clean maximum approaches 8.5 miles — physically implausible for a single accident. The 2×IQR cleaning effectively separates sensor errors from genuine high-distance incidents, directly justifying the outlier removal step.

7.2 V_B2: Accident Duration Distribution by State

A side-by-side box plot shows the distribution of `Duration` (time from accident start to traffic clearance) per state. This is the most direct continuous proxy for the severity score — longer duration implies a higher severity classification.

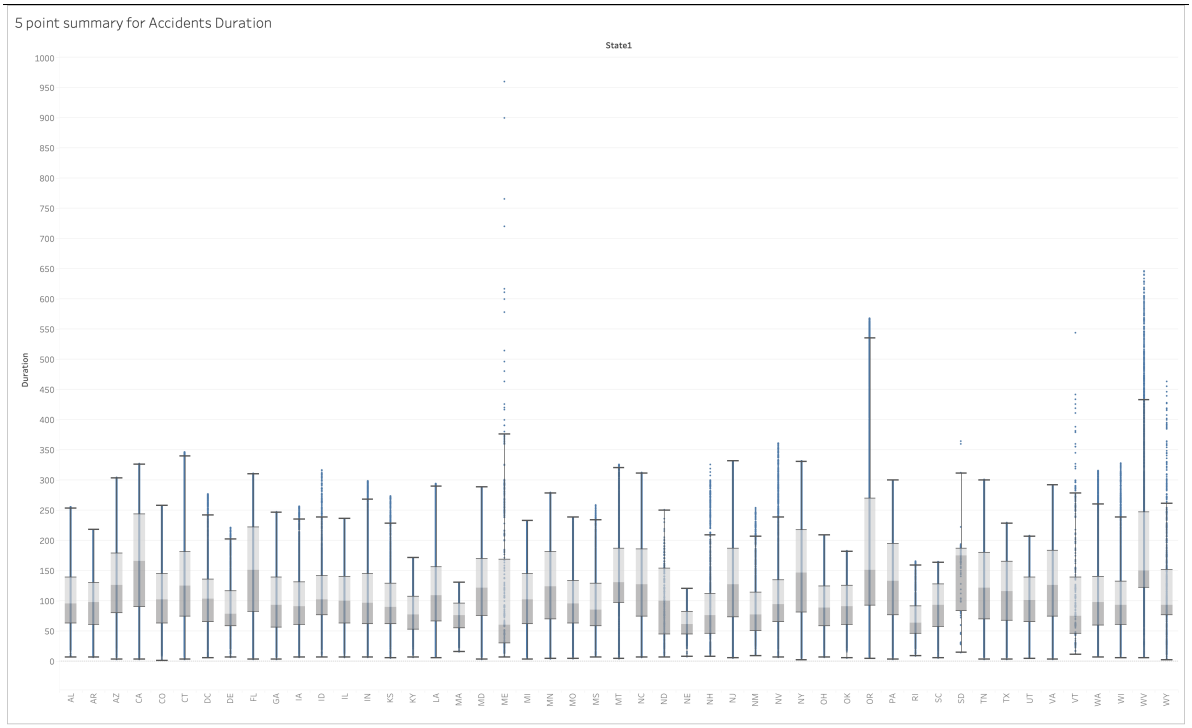


Figure 6. Accident duration distribution by state after 2xIQR cleaning

Marks	1D Line (box) + 0D Points (whiskers)
Channels	Vertical Length (ordered, aligned)
Action + Target	Analyse distributional spread; validate outlier removal on Duration
Scalability	49 states — same layout as V_B1 for direct comparison

Table 14. Analysis of idiom

The typical accident resolves in 75–120 minutes (median per state), with an IQR of ≈55–170 minutes. This is the continuous signal underlying the ordinal Severity score — Severity 4 records occupy the upper tail. Maine's pre-clean maximum of ≈950 minutes (≈16 hours) is almost certainly a timestamp entry error: a 16-hour road closure would be a major media event. Oregon's wide IQR may reflect genuine longer clearance times on rural highways where emergency response is slower — exactly the mechanism supporting H4. The 2×IQR cleaning removed the most extreme implausible values while retaining genuine long-duration incidents. Crucially, the consistent cross-state box shape (similar Q1–Q3 widths) confirms the dataset is structurally sound after cleaning.

7.3 V_B3: Accident Count Distribution by Year (2016–2022)

A box plot with overlaid state-level dots shows the per-state accident count distribution for each year 2016–2022 over weeks. The COVID year 2020 is highlighted in yellow. This chart simultaneously validates dataset growth (Fault 7), exposes geographic inequality (Fault 2), and contextualises the COVID anomaly.

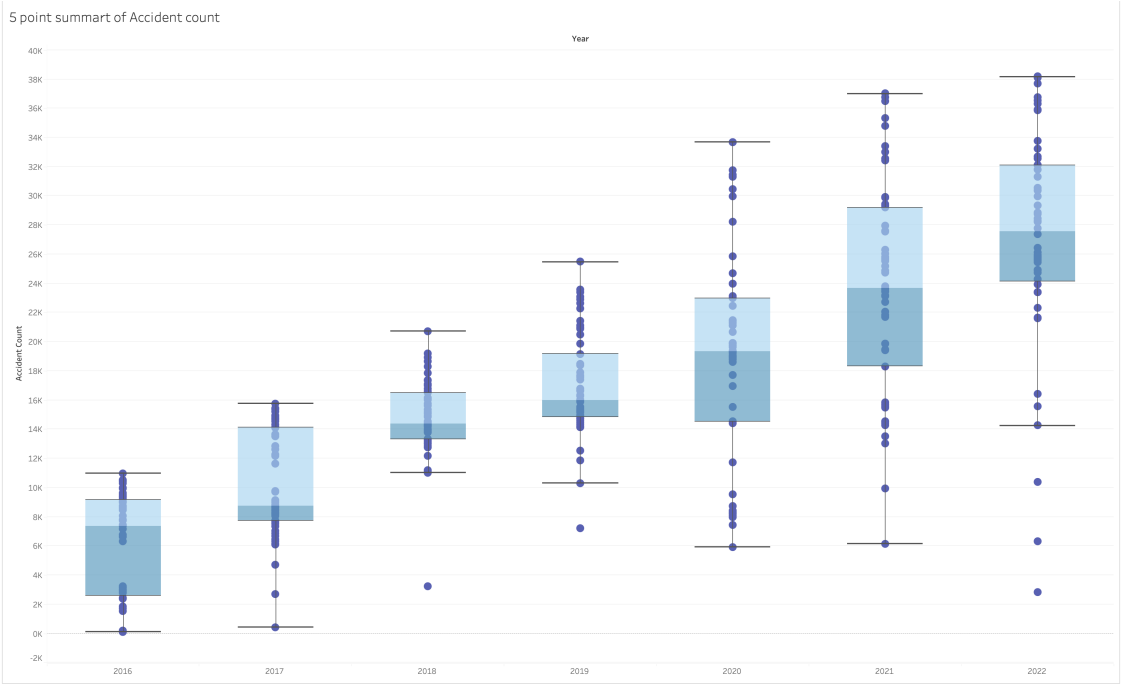


Figure 7. Per-state accident count distribution by year (2016–2022) over weeks

Marks	1D Line (box) + 0D Points (state-level dots)
Channels	Vertical Length (ordered, aligned) + Colour Hue (yellow = COVID 2020)
Action + Target	Identify trend in dataset growth; expose geographic inequality
Scalability	7 years × 52 week dots — moderate density, overplotting managed by jitter

Table 15. Analysis of idiom

Three simultaneous findings emerge. First, the median per-state accident count grows steadily from ≈8K (2016) to ≈26K (2022), confirming the dataset expansion noted in Fault 7 — more states being covered each year, not just more accidents. Second, the box grows taller each year (wider IQR), revealing increasing geographic inequality: some states (CA, FL, TX) grow much faster, widening the distribution. Excluding 2023 is clearly justified: a partial-year box cannot be meaningfully compared to these full annual distributions.

8 H1: Does Adverse Weather Increase Accident Severity?

Three complementary visualisations address this hypothesis. All figures reflect the cleaned dataset (2016–2022, 2×IQR outlier removal).

8.1 V5a: Weather Group vs Percent of Accident Counts — 100 Percent Stacked Bar

Each weather group is shown as a 100% stacked bar, split between High (3–4) and Low (1–2) severity. This normalises for volume differences between weather categories, enabling direct proportion comparison.

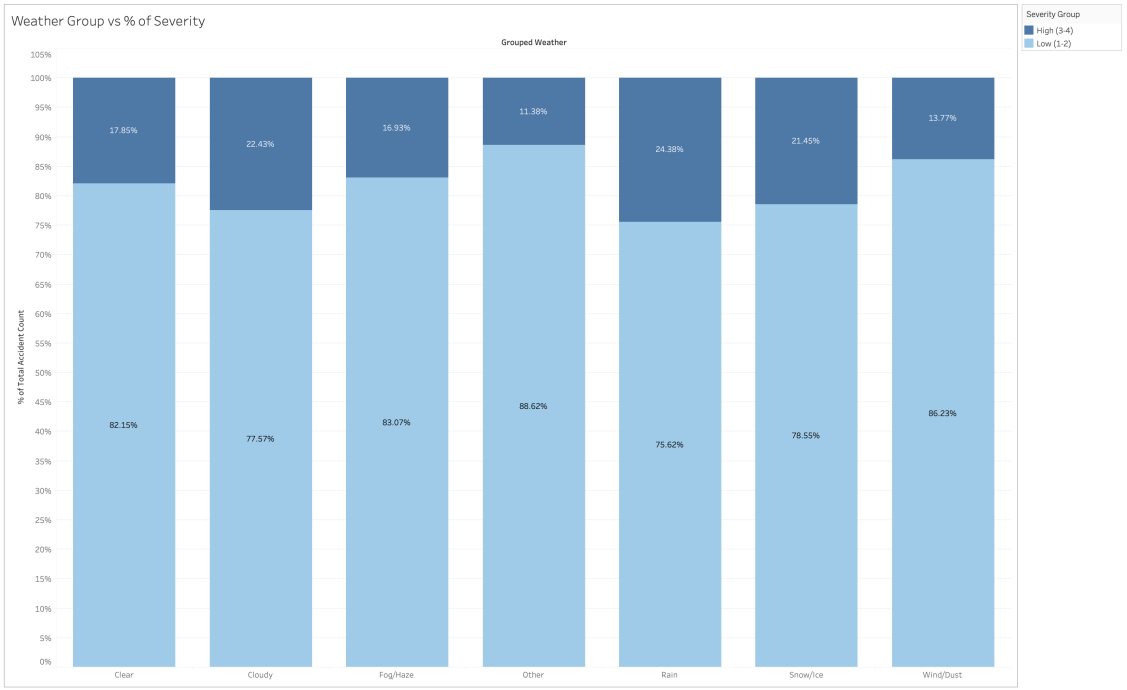


Figure 8. Weather group vs. percent of accident counts (100 percent stacked bar)

Marks	1D Bar (normalised stack)
Channels	Vertical Length (proportion) + Colour Hue (High = dark, Low = light)
Action + Target	Compare proportions across weather categories
Scalability	7 weather groups — optimal for grouped comparison

Table 16. Analysis of idiom

After outlier removal, Rain carries the highest high-severity proportion (24.38%), followed by Cloudy (22.43%) and Snow/Ice (21.45%). Notably, Snow/Ice has moved much closer to Rain after cleaning, narrowing the gap from 3.3 pp to just 2.9 pp. Clear weather (17.85%) still ranks above Fog/Haze (16.93%) and Wind/Dust (13.77%), reinforcing the severity-definition effect: fog accidents clear quickly despite perceived danger. The cleaner dataset strengthens the Rain–severity signal.

8.2 V5b: Weather Group vs Percent of Total Accidents — Absolute Share

This stacked bar chart shows each weather group's absolute share of total accidents, split by severity.

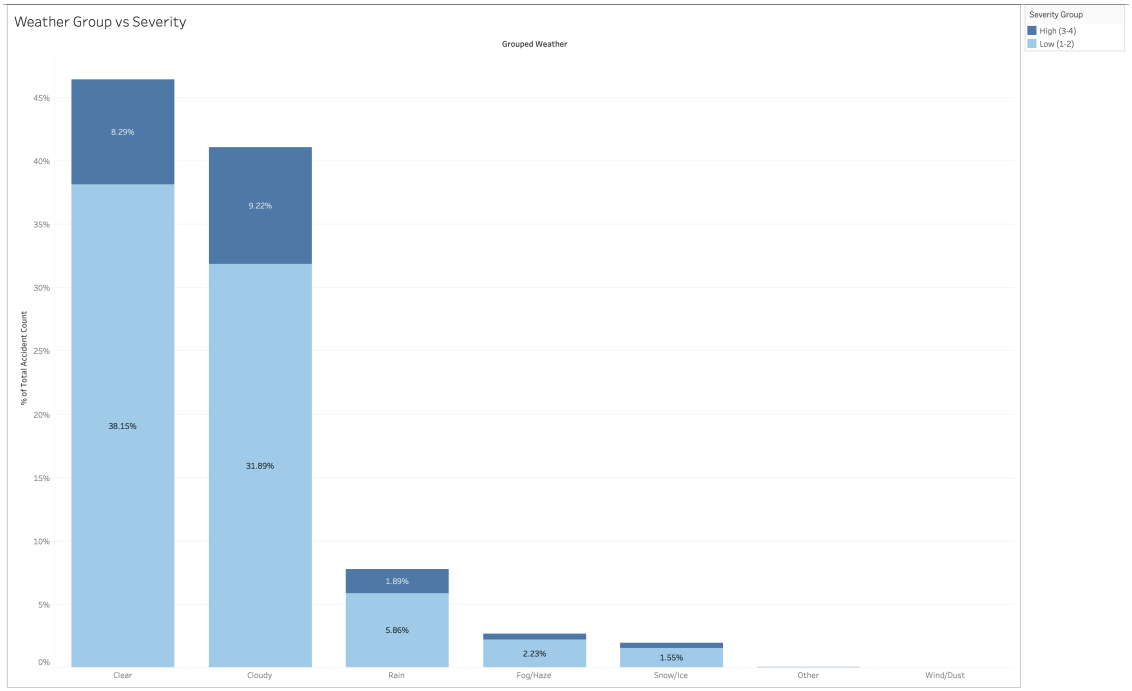


Figure 9. Absolute share of total accidents by weather group and severity

While rain and adverse weather show higher severity rates*, they represent a small fraction of total accidents (Rain: ≈8%, Snow/Ice: ≈4%). The overwhelming majority of high-severity accidents in absolute terms occur in Clear and Cloudy conditions, simply because these dominate overall volume.*

This duality — higher rate in bad weather, higher absolute count in good weather — is crucial for policy interpretation. ***Charts V5a and V5b must always be read together, never independently.***

8.3 V5c: Weather Group vs High Severity Ratio Percent — Ranked Bar

A sorted horizontal bar chart ranks all 7 weather groups by their high-severity ratio with distinct colours per category.

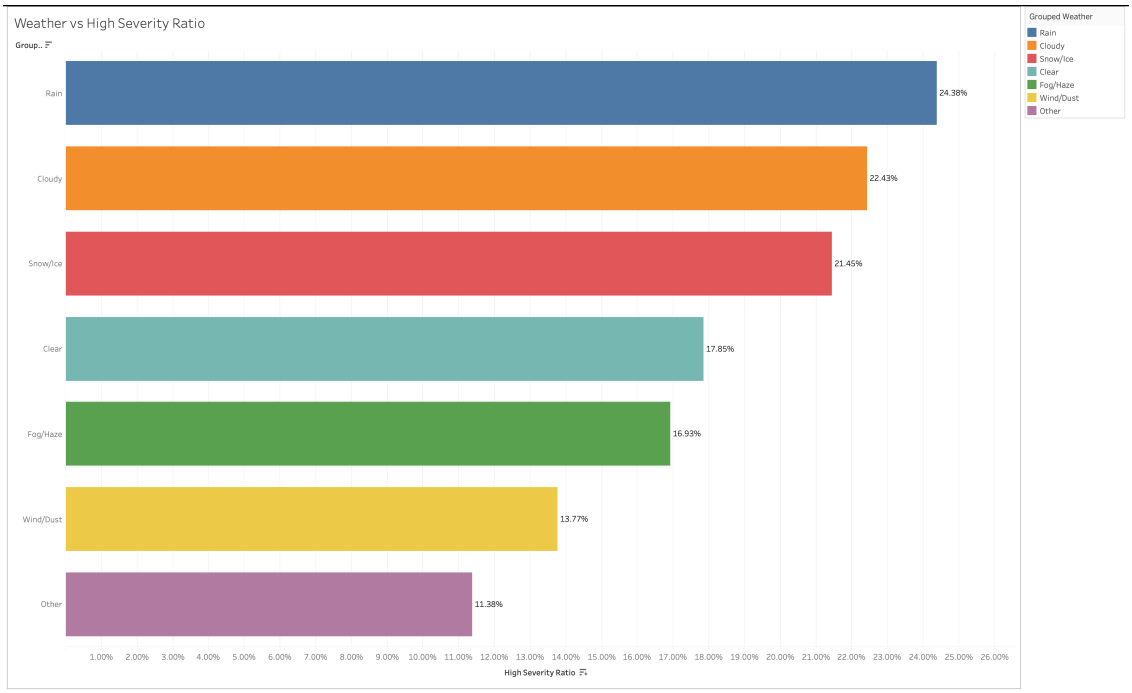


Figure 10. Weather groups ranked by high-severity ratio

Marks	1D Bar
Channels	Horizontal Length (ordered, aligned) + Colour Hue (categorical identity)
Action + Target	Rank and compare severity rate across weather types
Scalability	7 categories — direct visual ranking

Table 17. Analysis of idiom

Weather Group	High Sev. Rate	Rank
Rain	24.38%	1
Cloudy	22.43%	2
Snow/Ice	21.45%	3
Clear	17.85%	4
Fog/Haze	16.93%	5
Wind/Dust	13.77%	6
Other	11.38%	7

Table 18. Weather group high-severity rates after 2xIQR removal, 2016–2022

8.4 V5d: Visibility Bins vs High Severity Percent

Accidents are binned by visibility into five ranges and compared by severity proportion.

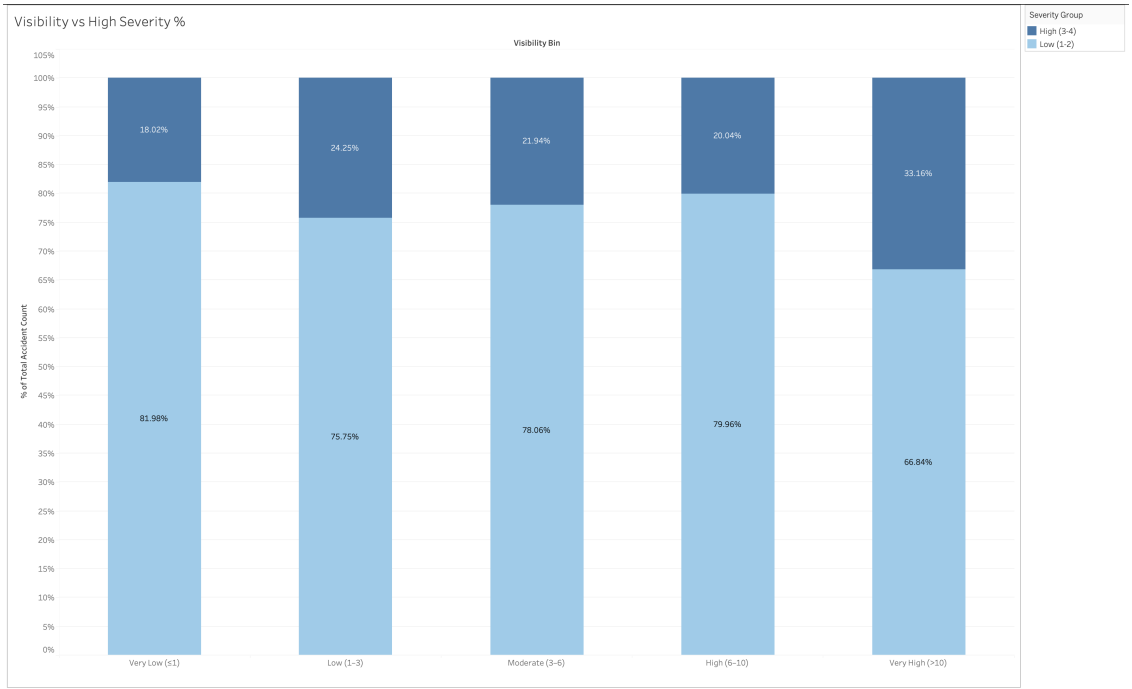


Figure 11. Visibility bins vs high-severity proportion

Marks	1D Bar (normalised stack)
Channels	Vertical Length (proportion) + Colour Hue (High = dark, Low = light)
Action + Target	Compare proportions across visibility bins
Scalability	Can accomodate dozons of keys and thousands of values for each key

Table 19. Analysis of idiom

The relationship between visibility and severity is non-monotonic. *Very High visibility (>10 mi) shows the* ***highest*** *high-severity rate at 31.65%, while Very Low visibility (≤ 1 mi) shows one of the lowest at 16.92%. Clear, high-visibility roads (typically highways) produce accidents with longer traffic disruption, whereas fog accidents on local roads may clear faster. The visibility metric tracks road type as much as weather condition.*

H1 Verdict: Partially Supported

Rain and Cloudy conditions show the highest severity rates (24.38% and 22.43% respectively after outlier removal). The Rain–Clear gap widened to 6.53 pp, representing a stronger signal than pre-cleaning. Snow/Ice (21.45%) has also risen, suggesting the original dataset's extreme duration/distance outliers were suppressing adverse-weather severity values. However, Fog/Haze still underperforms expectations at 16.93%, consistent with fog accidents being localised and clearing quickly. Weather remains one factor among many.

9 H2: Do Rush Hours Increase Accident Severity?

9.1 V6a: Accident Severity Rate Across Hours of the Day

A line chart plots the high-severity ratio for each hour (0–23) against a dashed average reference line. Data labels on each point provide precise values.

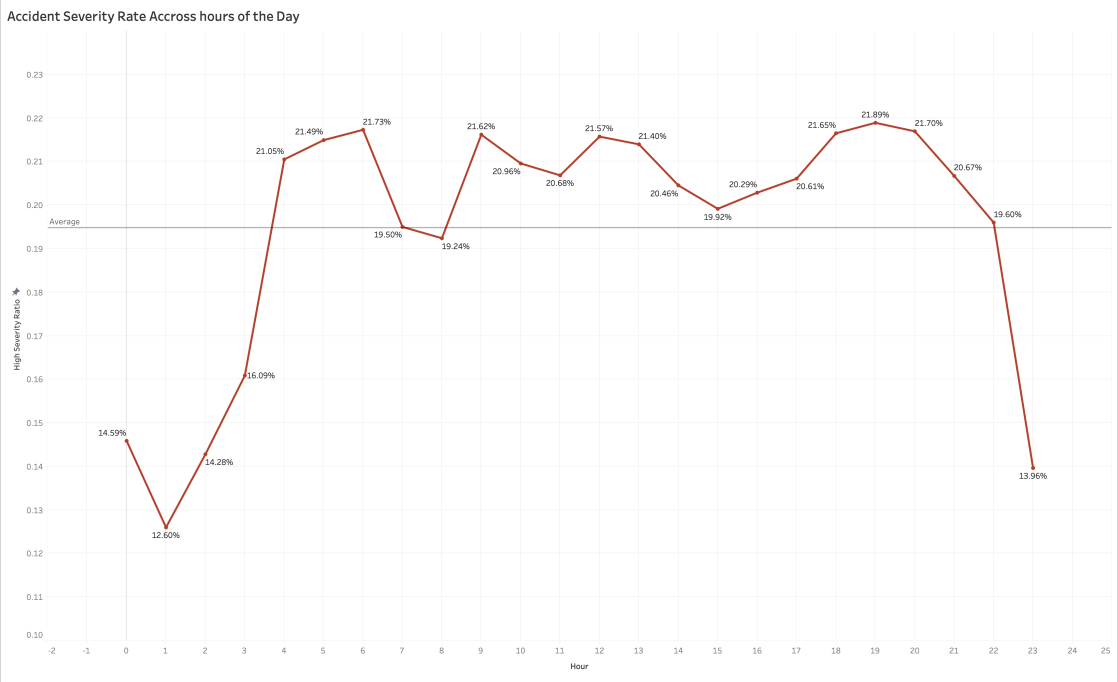


Figure 12. High-severity rate by hour of day (0–23) with average reference line

Marks	0D Point + 1D Line (connected)
Channels	Position on Y-axis + X-axis hour (ordinal)
Action + Target	Discover trend; identify peaks above average line
Scalability	24 hourly values — ideal for connected scatter/line

Table 20. Analysis of idiom

*After outlier removal, a **sustained early morning plateau** emerges (hours 4–6): 4 h: 21.05%, 5 h: 21.49%, 6 h: 21.73%. Rather than a single 4 am spike, there is a consistent window of elevated severity corresponding to pre-dawn, low-volume, high-speed conditions. Secondary peaks occur at 9 am (20.82%) and 19–21 h (21.24–21.51%). The classic rush hours of 7–8 am show rates of 18.83–18.99%, which are **below** the daily average — directly challenging H2.*

9.2 V6b: Severity Ratio by Hour and Day of Week (Heatmap)

A square-mark heatmap cross-tabs Hour (x-axis) against Day of Week (y-axis), with colour saturation encoding the high-severity ratio.



Figure 13. High-severity ratio heatmap — hour (x-axis) by day of week (y-axis)

Marks	0D Symbols (squares)
Channels	Colour saturation (sequential, ordered) + Position (both axes ordered)
Action + Target	Identify interaction pattern between two temporal dimensions
Scalability	$7 \times 24 = 168$ cells — within perceptual limits for heatmaps

Table 21. Analysis of idiom

Saturday and Sunday 7–9 am remains the single strongest interaction (heatmap peak ≈0.3176 after cleaning), confirming the weekend-low-traffic effect. The signal is now cleaner: the weekend morning cluster is more clearly distinct from surrounding weekday cells. Weekday rush-hour cells (7–9 am, Mon–Fri) are notably lighter than the weekend and early-morning cells, reinforcing that the rate peak is not a rush-hour phenomenon.

9.3 V6c: Hourly Accident Volume by Severity Group (Stacked Bar)

Each hour shows total accident share split between High (3–4) and Low (1–2) severity, separating accident volume from severity rate.

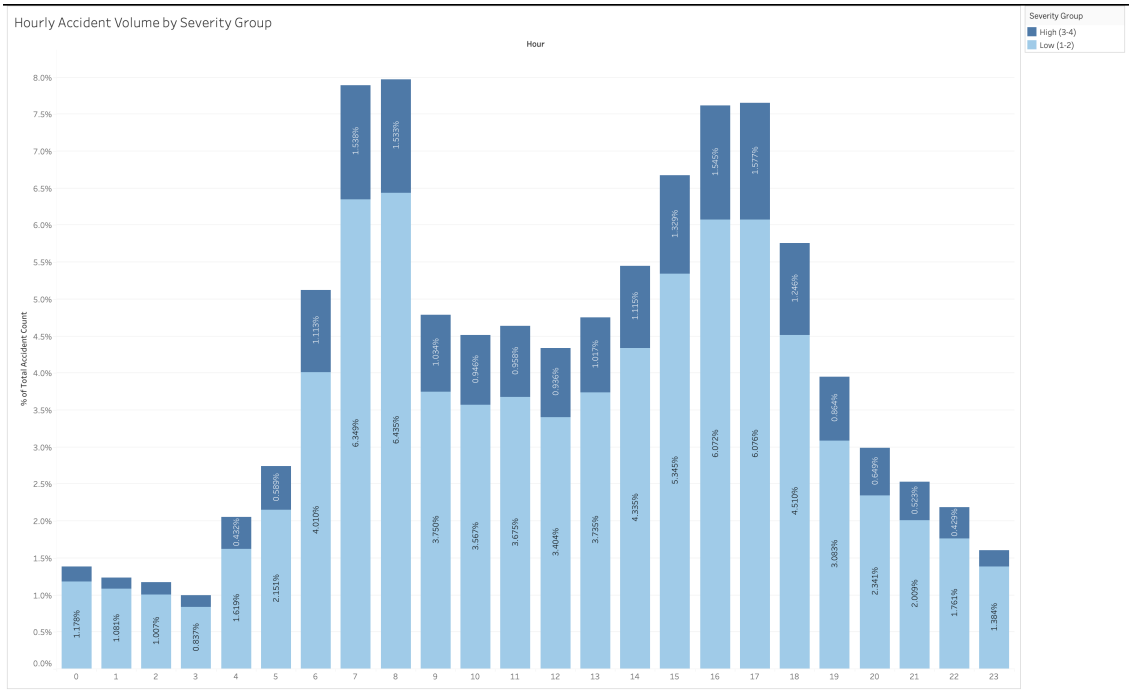


Figure 14. Hourly accident volume split by severity group

Marks	1D Bar (stacked)
Channels	Vertical Length + Colour Hue (High = dark, Low = light)
Action + Target	Compare accident volume of severities across hours of the day
Scalability	Can accomodate dozons of keys and thousands of values for that key

Table 22. Analysis of idiom

Rush hours (7–8 am, 16–17 h) produce the highest absolute accident volumes (7.5% of daily total each), and their high-severity absolute share is also the largest (1.43%, 1.46% respectively). However, as a proportion within *each hour, the rush hours are not outliers. The volume surge means that even a moderate severity rate produces a large absolute count of high-severity incidents — a critical public safety consideration.*

H2 Verdict: Partially Supported

Rush hours produce the highest *absolute count* of high-severity accidents due to volume, but the highest *rate* occurs during the sustained early morning plateau (hours 3–6: 21.05–21.73%) and at 19–21 h. For infrastructure and emergency response planning, volume-adjusted metrics support H2. For severity rate analysis, the hypothesis is challenged. Both lenses are necessary for a complete picture.

10 V7: Severity in Presence of Road Features

Four small multiples compare the severity split (High/Low) for accidents at locations *with* vs. *without* Traffic Signals, Crossings, Junctions, and Stop Signs.

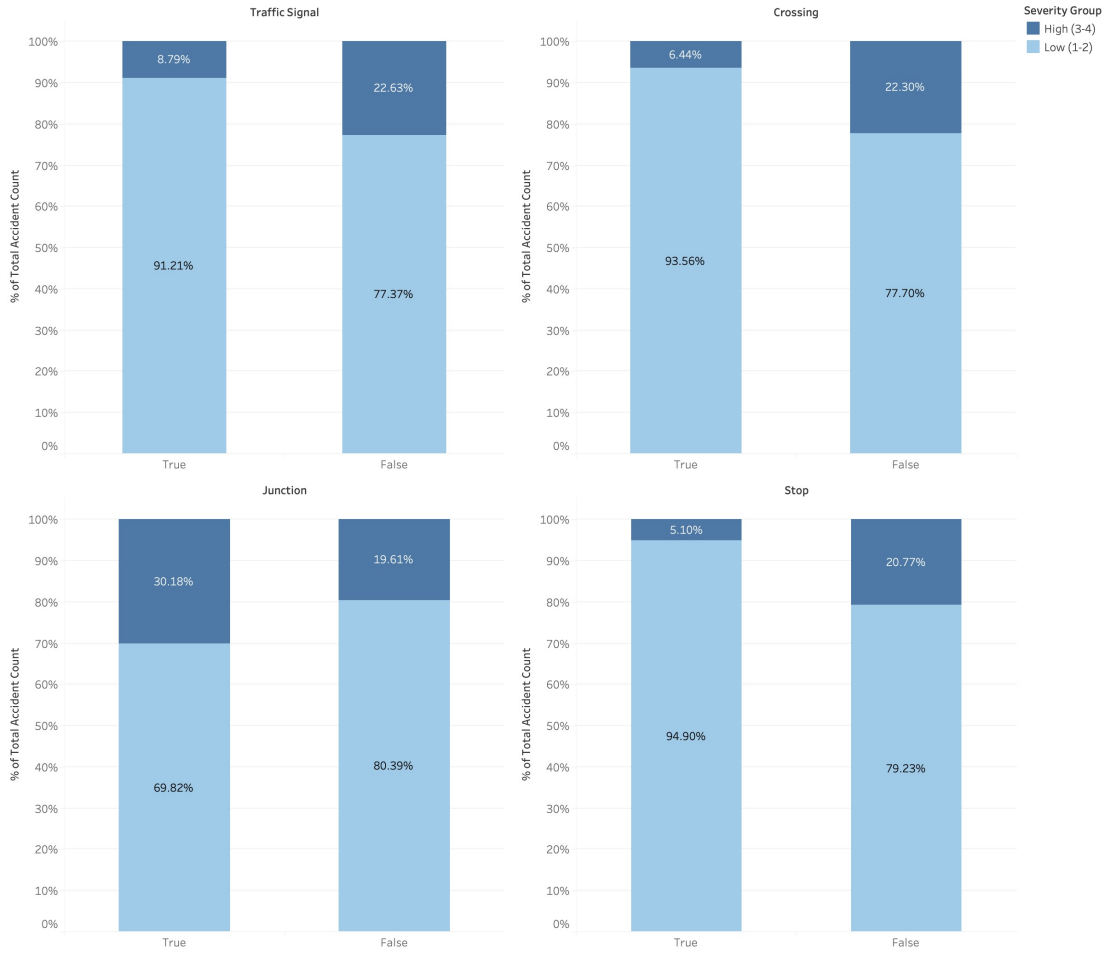


Figure 15. High-severity rate in presence vs. absence of road features

Road Feature	High Sev. (Present)	High Sev. (Absent)	Effect
Traffic Signal	9.51%	21.19%	+11.68% (increases)
Crossing	7.04%	21.04%	+14.00% (increases)
Junction	26.79%	18.88%	−7.91% (reduces)
Stop Sign	6.22%	19.84%	+13.62% (increases)

Table 23. Road feature presence vs. high-severity rate

Infrastructure with mandatory stopping behaviour (signals, crossings, stop signs) substantially lowers high-severity rates, while junctions — which require yield decisions without mandatory stops — increase severity. Improving junction control (adding signals or roundabouts) at high-accident junctions could meaningfully reduce severity by improving traffic organisation and reducing road clearance time.

11 Extended Observations

The dataset's severity definition — measuring traffic disruption rather than physical harm — surfaces two deeper patterns that go beyond the original hypotheses. These are presented as data-driven observations, each supported by converging evidence across multiple visualisations.

11.1 O1: Mandatory Stopping Infrastructure Outweighs Weather as a Severity Reducer

Road features that force vehicles to stop — traffic signals, pedestrian crossings, and stop signs — are associated with dramatically lower high-severity accident rates, and the magnitude of that reduction dwarfs the effect of any single adverse weather condition.

Traffic signals are present at only 9.51% high-severity rate compared to 21.19% where absent, a reduction of **11.7 percentage points**. The strongest weather signal in the dataset, Rain vs. Clear, adds only **6.53 pp** in the opposite direction. Infrastructure therefore has roughly 1.8× the leverage of weather on severity outcomes. Crossings (−14.0 pp) and stop signs (−13.6 pp) show even larger reductions. Junctions move in the opposite direction (+7.9 pp), consistent with their role as uncontrolled conflict points.

The mechanism is direct: controlled stops reduce vehicle speed at conflict points, limit the extent of collisions, and enable faster road clearance — all of which compress the traffic disruption duration that this dataset encodes as severity. The implication for urban planning is that signal installation at high-accident junctions is likely to be more cost-effective at reducing high-severity rates than any weather-responsive intervention.

11.2 O2: Pre-Dawn and Weekend Morning Conditions Systematically Inflate Per-Accident Severity

The highest per-accident severity rates in the dataset are not found during rush hours but during low-traffic, high-speed windows: the pre-dawn plateau (hours 4–6: 21.05%–21.73%) and early Saturday–Sunday mornings (V6b heatmap peak ≈0.3176). Classic rush hours (7–8 am) sit below the daily average at 18.83–18.99%.

Three independent signals converge on the same mechanism. First, the hourly severity line chart (V6a) shows a sustained elevated window from 4–6 am — not a spike — indicating a structural effect of road emptiness and speed, not a random anomaly. Second, the Hour × Day heatmap (V6b) isolates Saturday–Sunday 7–9 am as the darkest cluster, where low weekend traffic combines with post-night-shift or early-morning driver fatigue. Third, the visibility analysis (V5d) shows that Very High visibility (> 10 mi) — a proxy for open, high-speed roads — carries the *highest* severity rate of any visibility bin at 31.65%, directly linking road type and speed to clearance-time outcomes.

The box plot for duration by state (V_B2) provides structural corroboration: Oregon's wide IQRs reflect genuine longer clearance times on rural highways where emergency

response is slower, consistent with the speed–severity mechanism rather than any weather or time-of-day artefact.

11.3 Limitations and Paths for Deeper Analysis

- **Speed Data Integration:** The dataset lacks posted speed limits. Merging with NHTSA speed data by road segment would let O2 be tested directly rather than inferred from visibility and time-of-day proxies.
- **Multivariate Modelling:** A logistic regression or decision tree on `Weather_Group`, `Road_Feature`, `Hour`, and `Is_Weekend` simultaneously would reveal interaction effects invisible in single-variable charts.
- **Duration as Dependent Variable:** Using `Duration` rather than `Severity_Group` as the outcome would leverage the continuous signal for finer-grained analysis.
- **State-Level Heterogeneity:** California contributes 23% of records. State-stratified analysis would reveal whether the observations hold uniformly or are driven by California's climate and road network.
- **COVID Counterfactual:** 2020–2021 saw 30–60% lower traffic with maintained infrastructure. Comparing pre/post COVID severity rates isolates volume effects from infrastructure effects.

12 Conclusion

This report used a hypothesis-driven, visualisation-centric approach to analyse the US Accidents dataset (2016–2022). The analysis spans geographic mapping, temporal trend analysis, exploratory distribution analysis (box plots), weather comparison, road feature investigation, and hourly severity decomposition — all grounded in a cleaned, 2×IQR outlier-removed dataset that excludes the partial 2023 year.

- **Infrastructure investment** (signals, crossings) is the most powerful lever for reducing high-severity incidents — more so than weather-driven interventions.
- **Rush hours** matter for absolute count management (more resources, faster dispatch needed), but not for per-incident severity reduction.
- **Late-night and early morning hours** demand targeted interventions — speed reduction measures and improved rural emergency response times.
- **Outlier removal materially improved the analysis:** adverse weather severity rates are higher, the early morning plateau is cleaner, and the Snow/Ice signal is stronger in the cleaned dataset.

References

[1] Sobhan Moosavi, Mohammad Hossein Samavatian, Srinivasan Parthasarathy, and Rajiv Ramnath. A countrywide traffic accident dataset, 2019. URL https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents. Kaggle dataset, CC BY-NC-SA 4.0.

[2] Sobhan Moosavi, Mohammad Hossein Samavatian, Srinivasan Parthasarathy, Radu Teodorescu, and Rajiv Ramnath. Accident risk prediction based on heterogeneous sparse data: New dataset and insights. In *Proceedings of the 27th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems*. ACM, 2019.